Nidhi Kumari

+91 9934170930 — nidhixyz5kumari@gmail.com — LinkedIn — GitHub — Portfolio

Education

Degree	Institution	CGPA/Percentage	Year
M.S. in QMS (Applied Statistics & OR)	Indian Statistical Institute, Bengaluru	85.08%	2027
B.Sc. in Mathematics (Honours)	Institute of Science, Banaras Hindu University, Varanasi	8.53	2025

Internship Experience

Data Analysis Intern May 2026 – July 2026

CGI, Bengaluru

- Cleaned, preprocessed, and performed exploratory analysis on structured operational datasets using Python and Excel.
- Analyzed application error messages and failure logs to identify recurring failure patterns and support defect categorization.
- Applied statistical analysis to identify operational trends, extract business insights, and generate analytical reports.
- Collaborated with cross-functional teams to analyze operational trends and support data-driven decision-making.

Projects

Remaining Useful Life (RUL) Prediction for Aircraft Engines ***(Ongoing)***

- Performed exploratory data analysis on multivariate aircraft engine sensor data to examine sensor behaviour and degradation patterns across operational cycles.
- Analyzed relationships between sensor measurements and Remaining Useful Life (RUL), and refined the feature set by removing constant and non-informative variables.
- Engineered rolling mean and baseline deviation features for selected sensors to capture temporal degradation trends and deviations from normal operating conditions.
- Developing a feature-engineered dataset for subsequent model development and evaluation for predicting the Remaining Useful Life of aircraft engines.

Traveling Salesman Problem (TSP) Optimization

- Formulated the Traveling Salesman Problem as a Mixed Integer Linear Programming (MILP) model for optimal route planning.
- Implemented Gurobi and OR-Tools, and explored Warm Start and KNN Sparsification to improve computational efficiency and scalability.
- Evaluated optimization performance across multiple problem sizes using runtime, optimality gap, and solution quality.
- Analyzed the trade-offs between exact optimization and sparsification approaches for large-scale routing problems.

Supplier Risk Prediction using Machine Learning

- Developed a Logistic Regression model to predict supplier quality risk from structured supplier data.
- Performed data preprocessing, exploratory data analysis, and feature analysis to identify factors associated with supplier risk.
- Evaluated model performance using a confusion matrix, precision, recall, and F1-score to assess classification performance.
- Analyzed model predictions to identify potential high-risk suppliers and support data-driven supplier quality assessment.

Technical Skills

Programming: Python, SQL, Java, C

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Excel, Data Cleaning, Exploratory Data Analysis

Machine Learning: Scikit-learn, Logistic Regression, Linear Regression, Random Forest, XGBoost, Supervised Learning, Feature Engineering, Model Evaluation

Statistics: Hypothesis Testing, Statistical Inference, Regression Analysis, ANOVA

Operations Research: Linear Programming, Mixed Integer Linear Programming, Gurobi, OR-Tools, Optimization

Leadership Experience

Placement Representative Indian Statistical Institute, Bengaluru

- Coordinate with companies, recruiters, and placement coordinators to facilitate internship and placement activities.
- Manage interview scheduling, candidate communication, and recruitment-related coordination across multiple stakeholders.